

# FUNCTIONAL GROUPS IN ORGANIC CHEMISTRY

Functional groups are the characteristic groups in organic molecules that give them their reactivity. In the formulae below, R represents the rest of the molecule and X represents any halogen atom.

● Hydrocarbons ● Halogen-containing groups ● Oxygen-containing groups ● Nitrogen-containing groups ● Sulfur-containing groups ● Phosphorus-containing groups

|                                                                     |                                                            |                                                                             |                                                                             |                                                                                |                                                                       |                                                                              |                                                             |                                                                   |                                                                            |
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|                                                                     |                                                            |                                                                             |                                                                             |                                                                                |                                                                       |                                                                              |                                                             |                                                                   |                                                                            |
| <b>ALKANE</b><br>Naming: -ane<br>e.g. ethane                        | <b>ALKENE</b><br>Naming: -ene<br>e.g. ethene               | <b>ALKYNE</b><br>Naming: -yne<br>e.g. ethyne                                | <b>ARENE</b><br>Naming: -yl benzene<br>e.g. ethyl benzene                   | <b>HALOALKANE</b><br>Naming: halo-<br>e.g. chloroethane                        | <b>ALCOHOL</b><br>Naming: -ol<br>e.g. ethanol                         | <b>ALDEHYDE</b><br>Naming: -al<br>e.g. ethanal                               | <b>KETONE</b><br>Naming: -one<br>e.g. propanone             | <b>CARBOXYLIC ACID</b><br>Naming: -oic acid<br>e.g. ethanoic acid | <b>ACID ANHYDRIDE</b><br>Naming: -oic anhydride<br>e.g. ethanoic anhydride |
|                                                                     |                                                            |                                                                             |                                                                             |                                                                                |                                                                       |                                                                              |                                                             |                                                                   |                                                                            |
| <b>ACYL HALIDE</b><br>Naming: -oyl halide<br>e.g. ethanoyl chloride | <b>ESTER</b><br>Naming: -yl -oate<br>e.g. ethyl ethanoate  | <b>ETHER</b><br>Naming: -oxy -ane<br>e.g. methoxyethane                     | <b>EPOXIDE</b><br>Naming: -ene oxide<br>e.g. ethene oxide                   | <b>AMINE</b><br>Naming: -amine<br>e.g. ethanamine                              | <b>AMIDE</b><br>Naming: -amide<br>e.g. ethanamide                     | <b>NITRATE</b><br>Naming: -yl nitrate<br>e.g. ethyl nitrate                  | <b>NITRITE</b><br>Naming: -yl nitrite<br>e.g. ethyl nitrite | <b>NITRILE</b><br>Naming: -nitrile<br>e.g. ethanenitrile          | <b>NITRO</b><br>Naming: nitro-<br>e.g. nitromethane                        |
|                                                                     |                                                            |                                                                             |                                                                             |                                                                                |                                                                       |                                                                              |                                                             |                                                                   |                                                                            |
| <b>NITROSO</b><br>Naming: nitroso-<br>e.g. nitrosoethane            | <b>IMINE</b><br>Naming: -imine<br>e.g. ethanimine          | <b>IMIDE</b><br>Naming: -imide<br>e.g. succinimide                          | <b>AZIDE</b><br>Naming: -yl azide<br>e.g. phenylazide                       | <b>CYANATE</b><br>Naming: -yl cyanate<br>e.g. methyl cyanate                   | <b>ISOCYANATE</b><br>Naming: -yl isocyanate<br>e.g. methyl isocyanate | <b>AZO COMPOUND</b><br>Naming: azo-<br>e.g. azoethane                        | <b>THIOL</b><br>Naming: -thiol<br>e.g. methanethiol         | <b>SULFIDE</b><br>Naming: sulfide<br>e.g. dimethyl sulfide        | <b>DISULFIDE</b><br>Naming: disulfide<br>e.g. dimethyl disulfide           |
|                                                                     |                                                            |                                                                             |                                                                             |                                                                                |                                                                       |                                                                              |                                                             |                                                                   |                                                                            |
| <b>SULFOXIDE</b><br>Naming: sulfoxide<br>e.g. dimethyl sulfoxide    | <b>SULFONE</b><br>Naming: sulfone<br>e.g. dimethyl sulfone | <b>SULFINIC ACID</b><br>Naming: -sulfinic acid<br>e.g. benzenesulfinic acid | <b>SULFONIC ACID</b><br>Naming: -sulfonic acid<br>e.g. benzenesulfonic acid | <b>SULFONATE ESTER</b><br>Naming: -yl sulfonate<br>e.g. methylmethanesulfonate | <b>THIOCYANATE</b><br>Naming: thiocyanate<br>e.g. ethyl thiocyanate   | <b>ISOTHIOCYANATE</b><br>Naming: isothiocyanate<br>e.g. ethyl isothiocyanate | <b>THIAL</b><br>Naming: -thial<br>e.g. ethanethial          | <b>THIO KETONE</b><br>Naming: -thione<br>e.g. propanethione       | <b>PHOSPHINE</b><br>Naming: phosphine<br>e.g. methylphosphane              |



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Figure 5.2.1: Functional groups in organic chemistry. (CC BY-NC-ND, CompoundChem.com).

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